

Randomized and low-rank approximation

Super-exponential decay of the zero-permanent probability

Difficulty: extreme **Importance:** interesting to the community**Status:** Open

Rating rationale: Extreme difficulty reflects the unresolved passage from a fixed exponential rate to arbitrarily large rates, despite the recent exponential breakthrough. The question concerns a central cancellation phenomenon in combinatorial random-matrix theory, supporting community importance.

Problem statement

For each integer $n \geq 1$, let $A_n = (a_{ij}) \in \mathbb{R}^{n \times n}$ have independent entries with $\Pr(a_{ij} = 1) = \Pr(a_{ij} = -1) = 1/2$. Define

$$\text{per } A_n = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i, \sigma(i)}.$$

Is it true that, for every real $c > 0$, there exists N_c such that

$$\Pr(\text{per } A_n = 0) \leq e^{-cn} \quad \text{for every } n \geq N_c?$$

Equivalently, $\lim_{n \rightarrow \infty} \Pr(\text{per } A_n = 0)^{1/n} = 0$. This explicitly quantifies the source’s phrase “super exponentially small.”

Relevance and ratings

The question concerns cancellation in a multilinear function of a random matrix. It complements random-matrix invertibility theory and bounds for randomized permanent calculations, with connections to combinatorial probability. The large gap beyond the current exponential estimate justifies extreme difficulty.

References

- V. H. Vu, *Recent progress in combinatorial random matrix theory*, Probability Surveys 18 (2021), 179–200, Conjecture 6.12 ([primary paper](#); [journal](#)).
- Z. Hunter, M. Kwan and L. Sauerermann, *Exponential anticoncentration of the permanent* (2025), Theorem 1.1 and §1.4 ([primary manuscript](#); [author PDF](#)).

Status check

Hunter–Kwan–Sauerermann prove the displayed bound for one absolute positive constant, and explicitly identify Vu’s stronger assertion as a future direction. Their result resolves exponential growth of the permanent range, which is not counted separately. Searches for Vu’s conjecture, super-exponential permanent anticoncentration, and 2025–2026 found no complete resolution. The July 2026 [Ginibre-ensemble paper](#) treats Gaussian entries and does not establish the discrete Bernoulli assertion.

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Independently rechecked [Vu, Conjecture 6.12](#) and [Hunter–Kwan–Sauerermann, Theorem 1.1 and §1.4](#); the latter’s arXiv record still lists only its September 2025 version and explicitly leaves super-exponential decay open. The fixed-rate theorem is a weaker bound, so the status remains Open. Targeted later searches also checked the authors’ [March 2026 finite-field paper, Theorems 1.3–1.4](#) and the [July 2026 Gaussian paper](#): their different field or entry distribution does not prove this real Rademacher assertion. No complete resolution was located in this bounded search.